方案：医药领域药监问题内容挖掘

目标：通过 Python 进行文档内容挖掘，针对医药领域药监的 query，生成一份重点词汇。

技术和工具选择：

a. 文本处理和分词：
- jieba：用于中文文本的分词和关键词提取。
- 停用词字典：用于去除常见的无意义词汇。

b. 特征提取和关键词提取：
- sklearn：使用 feature_extraction.text 中的 TfidfVectorizer 进行特征提取和关键词挖掘，使用 tf-idf 算法进行特征权重计算。
- 中文主题模型训练库：例如 gensim 库，用于训练主题模型以挖掘文档的主题信息。

实施步骤：

a. 数据收集：收集医药领域药监相关的文档数据，可以从公开的文档、报告、数据库等来源获取。

b. 数据预处理：
- 对文档进行清洗和去除噪声，如去除 HTML 标签、特殊字符等。
- 使用 jieba 对文档进行分词，得到词语列表。
- 去除停用词，保留具有意义的词语。

c. 特征提取和关键词挖掘：
- 使用 TfidfVectorizer 对预处理后的文档进行特征提取，得到每个文档的特征向量。
- 结合特征向量计算词语的 tf-idf 权重，并提取出每个文档的关键词。

d. 主题挖掘：
- 使用 gensim 或其他中文主题模型训练库，对预处理后的文档进行主题模型训练。
- 根据模型结果，挖掘文档的主题信息，提取关键主题词。

结果展示：

根据关键词提取的结果，生成一份重点词汇报告或词云图，展示医药领域药监文档的关键词和主题信息。
模型优化和迭代：

根据实际结果的反馈和需求变化，对模型进行迭代优化，提高关键词提取和主题挖掘的准确性和效果。

初步范例代码如下（尚未完成训练）：

Gensim topic modeling and similarity analysis - windows ver 0.1

```
import pandas as pd
```

```
import pandas as pd
```

In [2]:

```
import regex
```

In [3]:

```
data = pd.read_csv('/Users/cmcqd.csv', low_memory=False)
```

In [4]:

```
data[data['Compound_Number'] == 'xxxx'].to_csv(r"/Users/subset.csv")
```

In [6]:

```
covid_data = pd.read_csv('/Users/subset.csv')
```

In [7]:

```
covid_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8252 entries, 0 to 8251
Data columns (total 20 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Unnamed: 0         8252 non-null   int64  
 1   Serial#            8252 non-null   int64  
 2   Query_ID           8252 non-null   int64  
 3   Data_Source        8252 non-null   object 
 4   CTD_Section        1385 non-null   object 
 5   Date               8220 non-null   object 
 6   Compound_Number    8252 non-null   object 
 7   Compound_Name      8252 non-null   object 
 8   Line               8252 non-null   object 
 9   Country            8249 non-null   object 
 10  Query_Number       8252 non-null   object 
 11  Clinical_Phase     152 non-null    float64
 12  Query_Key_Words    152 non-null    object 
 13  Secondary_Key_Word  0 non-null     float64
 14  Query              8252 non-null   object 
 15  GDMS_Link          152 non-null    object 
 16  Overlap            16 non-null     object 
 17  Follow Up          0 non-null      float64
 18  Spacy_Sents        8252 non-null   object 
 19  ner_text           8252 non-null   object 
dtypes: float64(3), int64(3), object(14)
memory usage: 1.3+ MB
```

In [8]:

```
#list = covid_data['ner_text'].tolist()
```

In [9]:

```
covid_data['ner_text'].head()
```

Out[9]:

```
0                                  [drug substance]
1    [batch, analysis, commercial-scale, batches, a...
2                                  [drug substance]
3    [Certificate of analysis, alternative supplier...
4                                    [information]
Name: ner_text, dtype: object
```

In [10]:

```
covid_data['ner_text2'] = covid_data['ner_text'].astype(str).apply(lambda x: x.replace('[','').replace(']','').replace(', ',' '))
```

In [11]:

```
covid_data['ner_text2'].head()
```

Out[11]:

```
0                                  drug substance
1    batch analysis commercial-scale batches altern...
2                                  drug substance
3    Certificate of analysis alternative supplier H...
4                                    information
Name: ner_text2, dtype: object
```

In [12]:

```
covid_data['ner_text3'] = covid_data['ner_text2'].astype(str).apply(lambda x: x.split())
```

In [13]:

```
covid_data['ner_text3'].head()
```

Out[13]:

```
0                                [drug, substance]
1    [batch, analysis, commercial-scale, batches, a...
2                                [drug, substance]
3    [Certificate, of, analysis, alternative, suppl...
4                                  [information]
Name: ner_text3, dtype: object
```

In [14]:

```
covid_data.to_csv(r"/Users/zhaod02/nlp/cmc/outputs/covid_subset.csv")
```

In [15]:

```
#covid_data.dropna(subset = ["ner_text3"], inplace=True) # drop those rows which have NaN value cells
```

In [16]:

```
#covid_data['ner_text2'] = covid_data['ner_text'].astype(str).apply(lambda x: x.replace('[','').replace(']',''))
```

In [17]:

```
import csv
```

In [18]:

```
corpus_raw = []
```

In [19]:

```
with open('/Users/subset.csv', encoding='utf-8') as csvfile:
    reader = csv.DictReader(csvfile)
    for row in reader:
        corpus_raw.append(row['Spacy_Sents'])
```

In [20]:

```
#ner_corpus = covid_data['ner_text3'].tolist()
```

In [21]:

```
corpus_raw[0:10]
```

Out[21]:

```
['For drug substance:\n1.',
 'Additional batch analysis for commercial-scale batches t
o support the alternative supplier of Nucleoside Triphosph
ates (GTP) (Hongene Biotech Corporation), since the provid
ed data is not sufficient (3.2.S.4.4).',
 'For drug substance:\n2.',
 'Certificate of analysis from alternative supplier (Honge
ne Biotech Corporation) for GTP (3.2.S.2.3).',
 'we need to present to clarify the following information;\
n1.',
 '\tStability study progress (information available) for b
atches DS FP5960, FP5961 & FT0774 , due to only 1 month was
 presented.',
 '2.\tJustification about change: Proteinase K incubation
time range adjusted from 10-15 minutes to ≥10 minutes, tech
nical support was not presented to determine that is no a c
ritical parameter for Andover and Grange Castle sites.',
 '3.\tHold Times protocol and validation report  section 3.
2.S.2.5, according to local regulation qualification proce
sses should be performed with commercial size batches, usin
g at least 3 consecutive batches; or failing that  technica
l support including reference that endorses you evaluation
 with only 1 batch FP5958.',
 'COFEPRIS is requesting additional information to present
 no later than 5-Aug-2022, but please provide ASAP:\nCMC In
formation:\n1.',
 '\tStability study - protocol and report with long term re
sults (9 M) for at least 3 consecutive commercial DP batche
s, because only 6M for 30mcg batches EW4564 and EW4565 were
 presented.']
```

In [22]:

```
len(corpus_raw)
```

Out[22]:

```
8252
```

In [23]:

```
#with open("/Users/ZHAOD02/nlp/cmc/outputs/cmc_corpus_ner_covid.txt", "w",
 encoding='utf-8') as external_file:
#       print(ner_corpus, file=external_file)
#       external_file.close()
```

In [24]:

```
import multiprocessing
```

In [25]:

```
num_cores = multiprocessing.cpu_count()
```

In [26]:

```
from gensim.models.doc2vec import TaggedDocument, Doc2Vec
```

In [27]:

```
from gensim.utils import simple_preprocess
```

In [28]:

```
tagged_corpus = []
```

In [29]:

```
for i, text in enumerate(corpus_raw):
    tagged_doc = TaggedDocument(simple_preprocess(text), [i])
    tagged_corpus.append(tagged_doc)
```

In [30]:

```
tagged_corpus [0:10]
```

Out[30]:

```
[TaggedDocument(words=['for', 'drug', 'substance'], tags=
[0]),
 TaggedDocument(words=['additional', 'batch', 'analysis',
'for', 'commercial', 'scale', 'batches', 'to', 'support',
'the', 'alternative', 'supplier', 'of', 'nucleoside', 'tri
phosphates', 'gtp', 'hongene', 'biotech', 'corporation', '
since', 'the', 'provided', 'data', 'is', 'not', 'sufficien
t'], tags=[1]),
 TaggedDocument(words=['for', 'drug', 'substance'], tags=
[2]),
 TaggedDocument(words=['certificate', 'of', 'analysis', 'f
rom', 'alternative', 'supplier', 'hongene', 'biotech', 'co
rporation', 'for', 'gtp'], tags=[3]),
 TaggedDocument(words=['we', 'need', 'to', 'present', 'to
', 'clarify', 'the', 'following', 'information'], tags=
[4]),
 TaggedDocument(words=['stability', 'study', 'progress', '
information', 'available', 'for', 'batches', 'ds', 'fp', '
fp', 'ft', 'due', 'to', 'only', 'month', 'was', 'presented
'], tags=[5]),
```

```
 TaggedDocument(words=['justification', 'about', 'change',
 'proteinase', 'incubation', 'time', 'range', 'adjusted',
'from', 'minutes', 'to', 'minutes', 'technical', 'support
', 'was', 'not', 'presented', 'to', 'determine', 'that', '
is', 'no', 'critical', 'parameter', 'for', 'andover', 'and
', 'grange', 'castle', 'sites'], tags=[6]),
 TaggedDocument(words=['hold', 'times', 'protocol', 'and',
 'validation', 'report', 'section', 'according', 'to', 'lo
cal', 'regulation', 'qualification', 'processes', 'should
', 'be', 'performed', 'with', 'commercial', 'size', 'batch
es', 'using', 'at', 'least', 'consecutive', 'batches', 'or
', 'failing', 'that', 'technical', 'support', 'including',
 'reference', 'that', 'endorses', 'you', 'evaluation', 'wi
th', 'only', 'batch', 'fp'], tags=[7]),
 TaggedDocument(words=['cofepris', 'is', 'requesting', 'ad
ditional', 'information', 'to', 'present', 'no', 'later',
'than', 'aug', 'but', 'please', 'provide', 'asap', 'cmc',
'information'], tags=[8]),
 TaggedDocument(words=['stability', 'study', 'protocol', '
and', 'report', 'with', 'long', 'term', 'results', 'for',
'at', 'least', 'consecutive', 'commercial', 'dp', 'batches
', 'because', 'only', 'for', 'mcg', 'batches', 'ew', 'and
', 'ew', 'were', 'presented'], tags=[9])]
```

In [31]:

```
#with open("/Users/zhaod02/nlp/cmc/outputs/cmc_tagged_corpus_ner_covid.txt
", "w", encoding='utf-8') as external_file:
#     print(tagged_corpus, file=external_file)
#     external_file.close()
```

In [32]:

```
import gensim
```

In [33]:

```
model = gensim.models.doc2vec.Doc2Vec(vector_size=200, min_count=2, wor
kers=num_cores, epochs=50)
```

In [34]:

```
model.build_vocab(tagged_corpus)
```

In [35]:

```
model.train(tagged_corpus, total_examples=model.corpus_count, epochs=mode
l.epochs)
```

In [36]:

```
from gensim.test.utils import get_tmpfile
```

In [37]:

```
fname = get_tmpfile("/Users/cmc_doc2vec_covid_model")
```

In [38]:

```
model.save(fname) #将训练生成的 doc2vec model 保存
```

In [39]:

```
#model = Doc2Vec.load(fname) #调用保存的 doc2vec model
```

In [40]:

```
model.infer_vector(simple_preprocess('certificate of analysis from alternative supplier'))
```

Out[40]:

```
array([ 1.38884143e-03,  7.04072416e-02, -7.80535936e-02,
 3.42288822e-01,
       -2.54947916e-02, -2.31186017e-01, -9.59850941e-03,
3.49732563e-02,
       -2.12594643e-01, -7.62163028e-02, -1.35040194e-01,
3.29780243e-02,
       -1.48635358e-01,  2.82723278e-01, -1.76377986e-02, -
1.44143522e-01,
       -7.55327716e-02, -1.26237288e-01, -2.15194263e-02, -
2.83947349e-01,
        4.49132584e-02, -9.72570851e-02,  5.20342402e-03, -
1.14983312e-04,
       -2.92807948e-02, -2.25303575e-01,  8.44862983e-02, -
1.10789500e-01,
       -3.03259522e-01,  1.10904900e-02,  8.28658789e-02, -
1.67344317e-01,
        2.09940612e-01,  9.72895846e-02,  3.28146294e-02, -
5.68919033e-02,
        9.93695576e-04, -3.15667968e-03, -1.70266647e-02,
4.27713431e-02,
       -1.61411062e-01, -6.70169592e-02, -1.45686522e-01,
6.83692843e-02,
        1.23256087e-01,  2.14130245e-02,  1.17330477e-02, -
1.23434462e-01,
        7.23212510e-02,  3.69168557e-02, -1.64456293e-01,
1.04933016e-01,
        1.68089032e-01, -8.76105949e-02,  6.33108541e-02, -
4.00185883e-02,
       -2.27251947e-02,  3.47853959e-01, -3.91931325e-01, -
7.49240145e-02,
        1.22325737e-02,  1.66361749e-01, -9.83972549e-02, -
1.84363976e-01,
       -1.51265249e-01,  9.55228880e-02,  5.87918945e-02,
3.11244816e-01,
       -5.96516803e-02,  1.31184146e-01, -1.55208319e-01,
1.11167438e-01,
        3.45519274e-01, -1.24266155e-01,  1.45483971e-01,
2.23566331e-02,
```

```
       3.94775301e-01,  1.71701074e-01, -3.11499298e-01, -
5.41842133e-02,
      -5.80012538e-02,  8.82394612e-02, -1.46322846e-01,
2.92607814e-01,
      -2.19493181e-01, -1.58116803e-01,  2.95257811e-02,
2.53709257e-01,
       3.82939540e-02,  7.26189390e-02,  1.74754634e-01,
1.96116567e-01,
      -1.59708702e-03,  2.08715230e-01,  3.64134043e-01,
3.66840094e-01,
      -1.63928419e-01, -5.58534265e-02,  1.71128422e-01, -
6.75249249e-02,
      -1.22234784e-01,  1.59772292e-01,  6.87472448e-02,
9.86643508e-02,
      -5.64143993e-02, -1.06553115e-01, -9.31252912e-02,
2.35485986e-01,
      -2.23976150e-01, -5.99955209e-02, -3.85713503e-02, -
1.95686504e-01,
      -1.12307280e-01,  1.10209947e-02,  3.33884731e-02,
7.54822716e-02,
      -1.13494573e-02, -8.43138248e-02,  5.63468374e-02, -
3.11325658e-02,
       7.13685201e-03,  3.98131996e-01,  3.88107039e-02, -
3.36758167e-01,
       1.25331387e-01,  4.00712609e-01,  6.34137914e-02,
1.18246287e-01,
      -2.32918054e-01, -8.91970843e-02,  9.66176391e-02, -
6.18381575e-02,
       1.85045768e-02,  1.43836706e-03, -9.11447406e-02, -
2.27665659e-02,
      -5.15087917e-02, -1.21162042e-01, -1.21101089e-01, -
3.59170675e-01,
       1.08253233e-01, -2.23018974e-01, -9.96760204e-02,
6.07236624e-02,
      -2.86437143e-02,  2.26154730e-01, -2.10444823e-01, -
2.68764440e-02,
      -1.45815150e-03,  5.38674369e-02, -6.58880081e-03,
1.00143179e-01,
       1.37838796e-02,  3.27423140e-02, -2.04007193e-01,
1.98506564e-01,
       1.01478092e-01,  1.34753272e-01, -1.89346477e-01,
1.63880274e-01,
      -5.39376289e-02,  1.74106523e-01,  1.04367652e-03, -
2.52371341e-01,
```

```
       3.19573507e-02, -1.06999859e-01, -5.47749139e-02, -
1.50358528e-01,
       2.42862422e-02,  1.80502385e-02, -3.02184016e-01, -
2.00591404e-02,
      -4.97908928e-02,  1.11722857e-01, -4.52869646e-02,
3.67451757e-02,
      -1.40065551e-01, -7.18081892e-02,  5.89310043e-02,
7.43194157e-03,
       4.07157615e-02,  2.00516090e-01, -5.19118086e-02, -
9.60510410e-03,
      -5.34132719e-02,  3.88140529e-02, -5.44384867e-02, -
1.99702196e-02,
       2.80397773e-01, -6.46358356e-02,  1.27423644e-01, -
4.44808597e-04,
       1.58821885e-02, -2.17273429e-01,  1.46210909e-01, -
4.05952558e-02,
      -7.38144740e-02, -2.09958464e-01, -1.31903088e-03, -
2.19680041e-01],
     dtype=float32)
```

In [41]:

```
from gensim.models import LdaModel
```

In [42]:

```
from gensim import corpora
```

In [43]:

```
#pip install spacy
```

In [44]:

```
#import spacy
```

In [45]:

```
#nlp = spacy.load("/Users/dong/NLP/scispacy_models/en_core_sci_md-0.4.0/en_
core_sci_md/en_core_sci_md-0.4.0")
```

In [46]:

```
#pip install regex
```

In [47]:

```
#pip install nltk
```

In [48]:

```
#from nltk.tokenize import RegexpTokenizer
```

In [49]:

```
#tokenizer = RegexpTokenizer(r'\w+|$[0-9.]+|\S+')
```

In [50]:

```
import numpy
```

In [51]:

```
import nltk
C:\Anaconda3\lib\site-packages\sklearn\feature_extraction
\image.py:167: DeprecationWarning: `np.int` is a deprecate
```

```
d alias for the builtin `int`. To silence this warning, use
 `int` by itself. Doing this will not modify any behavior a
nd is safe. When replacing `np.int`, you may wish to use e.
g. `np.int64` or `np.int32` to specify the precision. If yo
u wish to review your current use, check the release note l
ink for additional information.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  dtype=np.int):
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:35: DeprecationWarning: `np.float` is a deprecat
ed alias for the builtin `float`. To silence this warning,
use `float` by itself. Doing this will not modify any behav
ior and is safe. If you specifically wanted the numpy scala
r type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=np.finfo(np.float).eps,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:597: DeprecationWarning: `np.float` is a depreca
ted alias for the builtin `float`. To silence this warning,
 use `float` by itself. Doing this will not modify any beha
vior and is safe. If you specifically wanted the numpy scal
ar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=np.finfo(np.float).eps, copy_X=True, fit_path=True,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:836: DeprecationWarning: `np.float` is a depreca
ted alias for the builtin `float`. To silence this warning,
 use `float` by itself. Doing this will not modify any beha
vior and is safe. If you specifically wanted the numpy scal
ar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=np.finfo(np.float).eps, copy_X=True, fit_path=True,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:862: DeprecationWarning: `np.float` is a depreca
ted alias for the builtin `float`. To silence this warning,
 use `float` by itself. Doing this will not modify any beha
```

```
vior and is safe. If you specifically wanted the numpy scal
ar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=np.finfo(np.float).eps, positive=False):
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:1097: DeprecationWarning: `np.float` is a deprec
ated alias for the builtin `float`. To silence this warnin
g, use `float` by itself. Doing this will not modify any be
havior and is safe. If you specifically wanted the numpy sc
alar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  max_n_alphas=1000, n_jobs=None, eps=np.finfo(np.float).e
ps,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:1344: DeprecationWarning: `np.float` is a deprec
ated alias for the builtin `float`. To silence this warnin
g, use `float` by itself. Doing this will not modify any be
havior and is safe. If you specifically wanted the numpy sc
alar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  max_n_alphas=1000, n_jobs=None, eps=np.finfo(np.float).e
ps,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\least
_angle.py:1480: DeprecationWarning: `np.float` is a deprec
ated alias for the builtin `float`. To silence this warnin
g, use `float` by itself. Doing this will not modify any be
havior and is safe. If you specifically wanted the numpy sc
alar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=np.finfo(np.float).eps, copy_X=True, positive=Fals
e):
C:\Anaconda3\lib\site-packages\sklearn\linear_model\rando
mized_l1.py:152: DeprecationWarning: `np.float` is a depre
cated alias for the builtin `float`. To silence this warnin
g, use `float` by itself. Doing this will not modify any be
```

```
havior and is safe. If you specifically wanted the numpy sc
alar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  precompute=False, eps=np.finfo(np.float).eps,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\rando
mized_l1.py:320: DeprecationWarning: `np.float` is a depre
cated alias for the builtin `float`. To silence this warnin
g, use `float` by itself. Doing this will not modify any be
havior and is safe. If you specifically wanted the numpy sc
alar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=np.finfo(np.float).eps, random_state=None,
C:\Anaconda3\lib\site-packages\sklearn\linear_model\rando
mized_l1.py:580: DeprecationWarning: `np.float` is a depre
cated alias for the builtin `float`. To silence this warnin
g, use `float` by itself. Doing this will not modify any be
havior and is safe. If you specifically wanted the numpy sc
alar type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  eps=4 * np.finfo(np.float).eps, n_jobs=None,
```

In [52]:

```
from nltk import word_tokenize
```

In [53]:

```
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\ZHAOD02\AppData\Roaming\nltk_dat
a...
[nltk_data]   Package punkt is already up-to-date!
```

Out[53]:

```
True
```

In [54]:

```
covid_df = pd.read_csv('/Users/covid_subset.csv', low_memory=False)
```

In [55]:

```
ner_corpus = list(covid_df['ner_text2'])
```

In [56]:

```
ner_corpus[0:10]
```

Out[56]:

```
['drug substance',
```

```
 'batch analysis commercial-scale batches alternative Nucl
eoside Triphosphates GTP Hongene Biotech Corporation data
3.2.S.4.4',
 'drug substance',
 'Certificate of analysis alternative supplier Hongene Bio
tech Corporation GTP',
 'information',
 'Stability study progress information DS FP5960 FP5961 FT0
774 month',
 'Justification Proteinase K incubation minutes minutes te
chnical support Andover Grange Castle sites',
 'validation report section 3.2.S.2.5 local regulation com
mercial size consecutive batches technical support evaluat
ion batch FP5958',
 'COFEPRIS information ASAP Information',
 'Stability study protocol report long term results consecu
tive DP batches M batches EW4564 EW4565']
```

In [57]:

```
len(ner_corpus)
```

Out[57]:

```
8252
```

In [58]:

```
from nltk.tokenize import sent_tokenize, word_tokenize
```

In [59]:

```
import re
```

In [60]:

```
ner_strings = []
```

In [61]:

```
for x in ner_corpus:
    #x = re.sub('[^a-zA-Z]', '', str(x))
    ner_strings.append(str(x))
```

In [62]:

```
ner_strings[0:10]
```

Out[62]:

```
['drug substance',
 'batch analysis commercial-scale batches alternative Nucl
eoside Triphosphates GTP Hongene Biotech Corporation data
3.2.S.4.4',
 'drug substance',
 'Certificate of analysis alternative supplier Hongene Bio
tech Corporation GTP',
 'information',
 'Stability study progress information DS FP5960 FP5961 FT0
774 month',
```

```
 'Justification Proteinase K incubation minutes minutes te
chnical support Andover Grange Castle sites',
 'validation report section 3.2.S.2.5 local regulation com
mercial size consecutive batches technical support evaluat
ion batch FP5958',
 'COFEPRIS information ASAP Information',
 'Stability study protocol report long term results consecu
tive DP batches M batches EW4564 EW4565']
```

In [63]:

```
tokenized_ner = [nltk.word_tokenize(sentence) for sentence in ner_strings]
```

In [64]:

```
tokenized_ner[0:10]
```

Out[64]:

```
[['drug', 'substance'],
 ['batch',
  'analysis',
  'commercial-scale',
  'batches',
  'alternative',
  'Nucleoside',
  'Triphosphates',
  'GTP',
  'Hongene',
  'Biotech',
  'Corporation',
  'data',
  '3.2.S.4.4'],
 ['drug', 'substance'],
 ['Certificate',
  'of',
  'analysis',
  'alternative',
  'supplier',
  'Hongene',
  'Biotech',
  'Corporation',
  'GTP'],
 ['information'],
 ['Stability',
  'study',
  'progress',
  'information',
  'DS',
  'FP5960',
```

```
 'FP5961',
 'FT0774',
 'month'],
['Justification',
 'Proteinase',
 'K',
 'incubation',
 'minutes',
 'minutes',
 'technical',
 'support',
 'Andover',
 'Grange',
 'Castle',
 'sites'],
['validation',
 'report',
 'section',
 '3.2.S.2.5',
 'local',
 'regulation',
 'commercial',
 'size',
 'consecutive',
 'batches',
 'technical',
 'support',
 'evaluation',
 'batch',
 'FP5958'],
['COFEPRIS', 'information', 'ASAP', 'Information'],
['Stability',
 'study',
 'protocol',
 'report',
 'long',
 'term',
 'results',
 'consecutive',
 'DP',
 'batches',
 'M',
 'batches',
 'EW4564',
```

```
 'EW4565']]
```

In [65]:

```
len(tokenized_ner)
```

Out[65]:

```
8252
```

In [66]:

```
dictionary = corpora.Dictionary(tokenized_ner) #从预处理后的语料中生成字典
```

In [67]:

```
dictionary.token2id
```

Out[67]:

```
{'drug': 0,
 'substance': 1,
 '3.2.S.4.4': 2,
 'Biotech': 3,
 'Corporation': 4,
 'GTP': 5,
 'Hongene': 6,
 'Nucleoside': 7,
 'Triphosphates': 8,
 'alternative': 9,
 'analysis': 10,
 'batch': 11,
 'batches': 12,
 'commercial-scale': 13,
 'data': 14,
 'Certificate': 15,
 'of': 16,
 'supplier': 17,
 'information': 18,
 'DS': 19,
 'FP5960': 20,
 'FP5961': 21,
 'FT0774': 22,
 'Stability': 23,
 'month': 24,
 'progress': 25,
 'study': 26,
 'Andover': 27,
 'Castle': 28,
 'Grange': 29,
 'Justification': 30,
 'K': 31,
 'Proteinase': 32,
```

```
'incubation': 33,
'minutes': 34,
'sites': 35,
'support': 36,
'technical': 37,
'3.2.S.2.5': 38,
'FP5958': 39,
'commercial': 40,
'consecutive': 41,
'evaluation': 42,
'local': 43,
'regulation': 44,
'report': 45,
'section': 46,
'size': 47,
'validation': 48,
'ASAP': 49,
'COFEPRIS': 50,
'Information': 51,
'DP': 52,
'EW4564': 53,
'EW4565': 54,
'M': 55,
'long': 56,
'protocol': 57,
'results': 58,
'term': 59,
'FK5127': 60,
'FK5618': 61,
'mcg': 62,
'presentation': 63,
'stability': 64,
'CoAs': 65,
'Present': 66,
'manufacturing': 67,
'conditions': 68,
'diluted': 69,
'reconstitution': 70,
'condition': 71,
'sterility': 72,
'stress': 73,
'test': 74,
'RH': 75,
'Labeling': 76,
```

'documents': 77,
'LLD': 78,
'changes': 79,
'Leaflet': 80,
'patients/caregivers': 81,
'version': 82,
'Bacterial': 83,
'Endotoxin': 84,
'Pfizer': 85,
'Testing': 86,
'bacterial': 87,
'extraction': 88,
'lipid': 89,
'nanoparticles': 90,
'product': 91,
'qualification': 92,
'samples': 93,
'solvent': 94,
'testing': 95,
'administered': 96,
'sample': 97,
'tested': 98,
'Belgium': 99,
'CBER': 100,
'KS': 101,
'MI': 102,
'PGS-Kalamazoo': 103,
'PGS-McPherson': 104,
'PGS-Puurs': 105,
'Tris/Sucrose': 106,
'endotoxin': 107,
'untreated': 108,
'Drug': 109,
'FDA': 110,
'PGS-GC': 111,
'Product': 112,
'non-extracted': 113,
'plans': 114,
'state': 115,
'transfer': 116,
'A.': 117,
'Chesterfield': 118,
'DNA': 119,
'MAH': 120,

'bioburden': 121,
'circular': 122,
'contamination': 123,
'controls': 124,
'in-process': 125,
'linear': 126,
'plasmid': 127,
'processes': 128,
'template': 129,
'B.': 130,
'column/membrane': 131,
'control': 132,
'impurity': 133,
'intermediates': 134,
'process': 135,
'removal': 136,
'storage': 137,
'Leachable': 138,
'leachable': 139,
'months': 140,
'3.2.P.2.4': 141,
'long-term': 142,
'.': 143,
'A': 144,
'EL8723': 145,
'Thermal': 146,
'cycling': 147,
'stored': 148,
'ultrafrozen': 149,
'vials': 150,
'moving': 151,
'real': 152,
'thermal': 153,
'world': 154,
'case': 155,
'scenario': 156,
'bulk': 157,
'genealogy': 158,
'manufacturers': 159,
'Korea': 160,
'approval': 161,
'manufactured': 162,
'request': 163,
'(': 164,

```
')': 165,
'-CAP': 166,
'5': 167,
'Poly': 168,
'tail': 169,
'’': 170,
'EW6126': 171,
'Lot': 172,
'timepoint': 173,
'MFDS': 174,
'ICH': 175,
'lots': 176,
'timepoints': 177,
'differences': 178,
'&': 179,
'Comparison': 180,
'Fill': 181,
'Finish': 182,
'facilities': 183,
'LNP': 184,
'acceptance': 185,
'comparison': 186,
'criteria': 187,
'mibe': 188,
'production': 189,
'site': 190,
'Acceptance': 191,
'B': 192,
'parameters': 193,
'time': 194,
'Time': 195,
'filtration': 196,
'post': 197,
'reduction': 198,
'temperature': 199,
'acceptable': 200,
'hours': 201,
'impact': 202,
'quality': 203,
'Arzneimittel': 204,
'Authorization': 205,
'Baxter': 206,
'COVID-19': 207,
'Emergency': 208,
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'Exela': 209,
'Germany': 210,
'GmbH': 211,
'Hospira': 212,
'Inc': 213,
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'Manufacturers': 215,
'Mibe': 216,
'Oncology': 217,
'PBS/Sucrose': 218,
'PBS/sucrose': 219,
'Panama': 220,
'Pfizer-BioNTech': 221,
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'Pharmaceutical': 223,
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'Studies': 226,
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'Use': 229,
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'LLD/': 245,
'Peru': 246,
'SLE': 247,
'appropriate': 248,
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'missunderstanding': 251,
'moment': 252,

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'multiples': 253,
'review': 254,
'variations/events': 255,
'STN': 256,
'deviation': 257,
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'inquiries': 260,
'investigation': 261,
'material': 262,
'query': 263,
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'deviations': 284,
'prevent': 285,
'recurrence': 286,
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'sentence': 303,
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'number': 308,
'result': 309,
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'total': 311,
'RNA': 312,
'Queries': 313,
'donations': 314,
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'shelf-life': 317,
'EUL': 318,
'US': 319,
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'CRS': 322,
'Member': 323,
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'occurs': 326,
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'BNT162b2': 330,
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'Puurs': 332,
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'Q.2': 360,
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'Kaisha': 478,
'Private': 479,
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'point': 494,
'subvisible': 495,
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'ALC-0315': 526,
'DSPC': 527,
'excipient': 528,
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'BCV40420-A': 531,
'Data': 532,
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'PEARL': 535,
'Unofficial': 536,
'27apr': 537,
'Handover': 538,
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'OF': 542,
'RATIONALE': 543,
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'3.2.P.5.1': 548,
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'line': 550,
'3.2.P.2.5': 551,
'CCIT': 552,
'Stolberg': 553,
'seals': 554,
'Gerresheimer': 555,
'Petesburg': 556,
'St': 557,
'4': 558,
'Response': 559,
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'Allegopharma': 561,
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'Please': 564,
'Process': 565,
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'description': 570,
'table': 571,
'P5.2': 572,
'P5.3': 573,
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'Validation': 575,
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'Sterility': 580,
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'Catalent': 582,
'Delpharm': 583,
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'Initial': 585,
'Query': 586,
'a': 587,
'document': 588,
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'Comparability': 590,
'strength': 591,
'P5.1': 592,
'P5.4': 593,
'P8': 594,
'FK5132': 595,
'FM0703': 596,
'started': 597,
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'residual': 604,

'stranded': 605,
'µg': 606,
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'BioPharma': 609,
'Biopharma': 610,
'Division': 611,
'Erwin-Rentschler-Str': 612,
'Goldgrube': 613,
'Mainz': 614,
'Manufacturing': 615,
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'Rentschler': 617,
'SE': 618,
'Wyeth': 619,
'der': 620,
'Laupheim': 621,
'35041': 622,
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'Marburg': 625,
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'Science': 627,
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'Content': 631,
'Encapsulation': 632,
'Osmolality': 633,
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'expression': 636,
'in': 637,
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'sanitary': 639,
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"'s": 641,
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'Thickness': 648,

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'diameter': 655,
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'P.7.9': 657,
'Seal': 658,
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'SIP': 663,
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'and': 676,
'booth': 677,
'X': 678,
'efficacy': 679,
'ray': 680,
'vaccines': 681,
'MIV': 682,
'Products': 683,
'Therapeutic': 684,
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'Animal': 688,
'Free': 689,
'Origin': 690,
'CAPA': 691,
'Fisher': 692,
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'synthetic': 696,
'vegetable': 697,
'variation': 698,
'Approval': 699,
'HA': 700,
'country': 701,
'origin': 702,
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'CMC': 704,
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'Outerpack': 708,
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'complete': 710,
'Japan': 711,
'average': 712,
'maximum': 713,
'value': 714,
'error': 715,
'spec': 716,
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'consideration': 718,
'e.g': 719,
'percentage': 720,
'statistical': 721,
'correct': 722,
'll': 723,
'circular/linear': 724,
'plan': 725,
'Change': 726,
'Comirnaty': 727,
'Corminaty': 728,
'Suite': 729,
'administrative': 730,
'filing': 731,
'queries': 732,
'PCO': 733,
'record': 734,
'urgency': 735,
'Pfizer/BioNTech': 736,

'hour': 737,
'media': 738,
'simulations': 739,
'campaigning': 740,
'connect/disconnect': 741,
'intervention': 742,
'simulation': 743,
'unique': 744,
'18': 745,
'Comment': 746,
'Line': 747,
'associated': 748,
'interventions': 749,
'manipulations': 750,
'with': 751,
'IND': 752,
'communication': 753,
'item': 754,
'restate': 755,
'format': 756,
'relevant': 757,
'Meet': 758,
'dsRNA': 759,
'CoA': 760,
'old': 761,
'DP/DS': 762,
'Global': 763,
'Team': 764,
'released': 765,
'ship': 766,
'elemental': 767,
'impurities': 768,
'BNT162B2': 769,
'adjustment': 770,
'formula': 771,
'ingredients': 772,
'pH': 773,
'ingredient': 774,
'quantity': 775,
'Vitro': 776,
'assays': 777,
'health': 778,
'levels': 779,
'3.2.P.5.4': 780,

'BNT1': 781,
'Anagni': 782,
'S.R.L': 783,
'analyzed': 784,
'bacth': 785,
'1': 786,
'No': 787,
'affidavit': 788,
'annex': 789,
'CMN': 790,
'Cie': 791,
'GC': 792,
'Merck': 793,
'Residual': 794,
'Solvents': 795,
'robustness': 796,
'LDPE': 797,
'bags': 798,
'appearance': 799,
'identification': 800,
'63017': 801,
'EU': 802,
'Inc.': 803,
'Laboratories': 804,
'MO': 805,
'OCABR': 806,
'Parkway': 807,
'St.Louis': 808,
'West': 809,
'FP8290': 810,
'Management': 811,
'allocation': 812,
'countries': 813,
'CTD': 814,
'IPC': 815,
'IPT-C': 816,
'assigned': 817,
'filters': 818,
'intermediate': 819,
'measured': 820,
'FP0377': 821,
'Manufacturer': 822,
'TFDA': 823,
'Translation': 824,

'authority': 825,
'central': 826,
'competent': 827,
'manufacturer': 828,
'BOH': 829,
'Tris/sucrose': 830,
'commitment': 831,
'Derivation': 832,
'Q5D': 833,
'Zagreb': 834,
'banks': 835,
'biotechnological/biological': 836,
'cell': 837,
'introduction/establishment': 838,
'substrates': 839,
'Biomay': 840,
'GH': 841,
'Kundl': 842,
'working': 843,
'In-Process': 844,
'Development': 845,
'P2.3': 846,
'Zeta': 847,
'charge': 848,
'potential': 849,
'surface': 850,
'Results': 851,
'difference': 852,
'observed': 853,
'variability': 854,
'Obligaion': 855,
'EL8983': 856,
'bridge': 857,
'heightened': 858,
'side': 859,
'3.2.P.3.5': 860,
'Malaysian': 861,
'P3.4': 862,
'Repligen': 863,
'Pall': 864,
'Santorius': 865,
'applications': 866,
'amount': 867,
'ET3045': 868,

```
'EX2405': 869,
'theamount': 870,
'PPQ3': 871,
'exckyded': 872,
'fixed': 873,
'FJ5682': 874,
'bracket': 875,
'compatibility': 876,
'condiucted': 877,
'deadline': 878,
'Anvisa': 879,
'Brazilian': 880,
'Inform': 881,
'area': 882,
'market': 883,
'mechanisms': 884,
'releasing': 885,
'AG': 886,
'Forward': 887,
'inclusion': 888,
'1.7.3': 889,
'3.2.S.2.1': 890,
'API': 891,
'APIs': 892,
'Certificates': 893,
'Compliance': 894,
'GMP': 895,
'Ultrafiltration/Diafiltration': 896,
'certification': 897,
'WSL10': 898,
'bank': 899,
'compliant': 900,
'manufacture': 901,
'non-CTD': 902,
'*': 903,
'Chesterfiled': 904,
'St.Loius': 905,
'The': 906,
'regulatory': 907,
'APRIL': 908,
'FN4073': 909,
'FN5988': 910,
'FN9606': 911,
'FP0362': 912,
```

```
'FP8544': 913,
'FR4267': 914,
'FR8392': 915,
'RESOLUTION': 916,
'approved': 917,
'biological': 918,
'materials': 919,
'narrowing': 920,
'publication': 921,
'replacement': 922,
'standard': 923,
'starting/intermediate': 924,
'10mcg': 925,
'Serbia': 926,
'microbiological': 927,
'verification': 928,
'procedures': 929,
'validation/verification': 930,
'Paediatric': 931,
'part': 932,
'closure': 933,
'verbally': 934,
'EVA': 935,
'containers': 936,
'2.25ml': 937,
'September/': 938,
'BNT': 939,
'FG4159': 940,
'leaflet': 941,
'FG0225': 942,
'Falstrich/Pamela': 943,
'Siwik': 944,
'William': 945,
'colleagues': 946,
'values': 947,
'agenda': 948,
'FG0222': 949,
'GF0222': 950,
'attend': 951,
'evening': 952,
'Host': 953,
'Linear': 954,
'Template': 955,
'figures': 956,
```

'檢測結果包括在確效結果 Range': 957,
'NMT1': 958,
'kanamycin': 959,
'-Cap': 960,
'IP-RP-HPLC': 961,
'Length*': 962,
'RP-HPLC': 963,
'Tail': 964,
'Materials': 965,
'CG': 966,
'Chemikalien': 967,
'advise': 968,
'audit': 969,
'confirmation': 970,
'please': 971,
'suppliers': 972,
'e-mail': 973,
'EUA': 974,
'team': 975,
'M0000050': 976,
'discrepancy': 977,
'duration': 978,
'increased': 979,
'incubated': 980,
'pre-incubation': 981,
'CTD3.2.A.3.7.3': 982,
'PMDA': 983,
'IM': 984,
'RTU': 985,
'modified': 986,
'partial': 987,
'pediatric': 988,
'insoluble': 989,
'year': 990,
'year/month': 991,
'BioPHarma': 992,
'Question': 993,
'USA': 994,
'linearized': 995,
'templates': 996,
'GCMC': 997,
'right': 998,
'ALC-159': 999,

```
 ...}
```

In [68]:

```
#将字典另存为cmcdict_covid.txt 文件
with open("/Users/cmcdict_covid.txt", "w", encoding ='utf-8') as external_fil
e:
    print(dictionary.token2id, file=external_file)
    external_file.close()
```

In [69]:

```
corpus_bow = [dictionary.doc2bow(text) for text in tokenized_ner]
```

In [70]:

```
corpus_bow[0:10]
```

Out[70]:

```
[[(0, 1), (1, 1)],
 [(2, 1),
  (3, 1),
  (4, 1),
  (5, 1),
  (6, 1),
  (7, 1),
  (8, 1),
  (9, 1),
  (10, 1),
  (11, 1),
  (12, 1),
  (13, 1),
  (14, 1)],
 [(0, 1), (1, 1)],
 [(3, 1), (4, 1), (5, 1), (6, 1), (9, 1), (10, 1), (15, 1),
(16, 1), (17, 1)],
 [(18, 1)],
 [(18, 1),
  (19, 1),
  (20, 1),
  (21, 1),
  (22, 1),
  (23, 1),
  (24, 1),
  (25, 1),
  (26, 1)],
 [(27, 1),
  (28, 1),
  (29, 1),
  (30, 1),
  (31, 1),
```

```
 (32, 1),
 (33, 1),
 (34, 2),
 (35, 1),
 (36, 1),
 (37, 1)],
[(11, 1),
 (12, 1),
 (36, 1),
 (37, 1),
 (38, 1),
 (39, 1),
 (40, 1),
 (41, 1),
 (42, 1),
 (43, 1),
 (44, 1),
 (45, 1),
 (46, 1),
 (47, 1),
 (48, 1)],
[(18, 1), (49, 1), (50, 1), (51, 1)],
[(12, 2),
 (23, 1),
 (26, 1),
 (41, 1),
 (45, 1),
 (52, 1),
 (53, 1),
 (54, 1),
 (55, 1),
 (56, 1),
 (57, 1),
 (58, 1),
 (59, 1)]]
```

In [71]:

```
#corpora.MmCorpus.serialize('/Users/zhaod02/nlp/cmc/outputs/cmc_doc2bow_corpus_covid.mm', corpus_bow) #将 doc2bow 语料库保存到本地
```

In [72]:

```
from gensim import models
```

In [73]:

```
tfidf = models.TfidfModel(corpus_bow) #TF-IDF vector conversion of bow corpus
```

In [74]:

```
for document in tfidf[corpus_bow][0:10]:
    print(document)
```

```
[(0, 0.6166172370504105), (1, 0.7872630964120685)]
[(2, 0.3752983932331141), (3, 0.2909165343917428), (4, 0.2
98447423267113), (5, 0.3074937672168333), (6, 0.2815560390
4408144), (7, 0.3752983932331141), (8, 0.406545844629458
3), (9, 0.23804966804751893), (10, 0.16339818518362556),
(11, 0.1259393696447917), (12, 0.11698238259879762), (13,
0.2909165343917428), (14, 0.089154639950902)]
[(0, 0.6166172370504105), (1, 0.7872630964120685)]
[(3, 0.3759718299517681), (4, 0.38570452554281215), (5, 0.
3973957499562361), (6, 0.3638746056655188), (9, 0.30764827
273350376), (10, 0.21117092853705208), (15, 0.375971829951
7681), (16, 0.24845998230707153), (17, 0.277887190828538)]
[(18, 1.0)]
[(18, 0.1532617345582077), (19, 0.1679592924489435), (20,
0.47185671340672997), (21, 0.47185671340672997), (22, 0.47
185671340672997), (23, 0.22736191428293404), (24, 0.269305
2082992778), (25, 0.3463924724058612), (26, 0.190125649548
6447)]
[(27, 0.19395064515514396), (28, 0.3369303258687541), (29,
 0.3369303258687541), (30, 0.2254256240385754), (31, 0.238
71867262717977), (32, 0.30112975316350654), (33, 0.2730524
1478423714), (34, 0.5156571124793219), (35, 0.167665950892
9731), (36, 0.30112975316350654), (37, 0.2785780594201583
4)]
[(11, 0.13058033690462328), (12, 0.12129327766798353), (3
6, 0.3188251603330654), (37, 0.2949482524620549), (38, 0.2
83899016063205), (39, 0.42152738661960376), (40, 0.1814709
3600250577), (41, 0.2949482524620549), (42, 0.216703584960
91266), (43, 0.3053783831541289), (44, 0.370176273476334
6), (45, 0.19401007694420103), (46, 0.1876339416163901),
(47, 0.18430463979980974), (48, 0.13160972740208193)]
[(18, 0.22290992977964666), (49, 0.6335384761921833), (50,
 0.6335384761921833), (51, 0.3841473175408176)]
[(12, 0.2866135038296881), (23, 0.23997350536751194), (26,
 0.20067177357455968), (41, 0.3484783069264319), (45, 0.22
922089748227875), (52, 0.1716067951264871), (53, 0.3708700
972011016), (54, 0.37668862639797573), (55, 0.295710235261
94407), (56, 0.2842433617482809), (57, 0.2533386079056858
5), (58, 0.1606473519811842), (59, 0.2808953700467789)]
```

In [75]:

```
from gensim.models import LdaModel
```

In [76]:

```
ldamodel = LdaModel(corpus=corpus_bow, num_topics=10, id2word=dictionary)
```

In [77]:

```
ldamodel.show_topics()[0:10]
```

Out[77]:

```
[(0,
  '0.041*"batch" + 0.038*"batches" + 0.031*"release" + 0.028*"DS" + 0.024*"lot" + 0.023*"data" + 0.023*"DP" + 0.020*"process" + 0.019*"CTM" + 0.018*"manufacturing"'),
 (1,
  '0.053*"modRNA" + 0.033*"RNA" + 0.023*"DS" + 0.019*"drug" + 0.019*"Lot" + 0.018*"products" + 0.017*"integrity" + 0.017*"data" + 0.015*"3" + 0.015*"BNT162b2"'),
 (2,
  '0.028*"assay" + 0.028*"development" + 0.026*"storage" + 0.025*"data" + 0.021*"containers" + 0.018*"information" + 0.018*"protein" + 0.017*"validation" + 0.017*"short-term" + 0.016*"lots"'),
 (3,
  '0.075*"stability" + 0.055*"studies" + 0.049*"data" + 0.026*"administration" + 0.019*"clinical" + 0.018*"storage" + 0.017*"product" + 0.017*"results" + 0.016*"preparation" + 0.011*"capped"'),
 (4,
  '0.151*"nan" + 0.023*"validation" + 0.016*"performed" + 0.015*"duration" + 0.014*"Protocol" + 0.013*"equipment" + 0.013*"classification" + 0.013*"diagrams" + 0.013*"facility" + 0.013*"Phase"'),
 (5,
  '0.047*"impurities" + 0.025*"content" + 0.023*"controlled" + 0.021*"process" + 0.018*"vials" + 0.018*"RNA" + 0.014*"FEI" + 0.014*"testing" + 0.012*"ALC-0315" + 0.012*"site"'),
 (6,
  '0.058*"drug" + 0.049*"Module" + 0.029*"product" + 0.028*"substance" + 0.024*"sections" + 0.017*"preparation" + 0.017*"pH" + 0.017*"locations" + 0.017*"RFI1" + 0.015*"bioburden"'),
 (7,
  '0.051*"tests" + 0.029*"manufacturing" + 0.029*"site" + 0.026*"DP" + 0.022*"information" + 0.021*"West" + 0.021*"Parkway" + 0.021*"DS" + 0.019*"dose" + 0.018*"controls"'),
 (8,
```

```
  '0.048*"criteria" + 0.035*"testing" + 0.034*"product" +
0.033*"clinical" + 0.030*"tests" + 0.028*"manufacturing" +
 0.023*"acceptance" + 0.021*"DS" + 0.020*"data" + 0.019*"D
P"'),
 (9,
  '0.036*"experience" + 0.024*"operations" + 0.024*"assess
ed" + 0.023*"Chesterfield" + 0.021*"clinical" + 0.020*"eva
luation" + 0.019*"criteria" + 0.017*"DNA" + 0.014*"trial"
+ 0.014*"potency"')]
```

In [78]:

```
#pip install pyldavis
```

In [79]:

```
import pyLDAvis
import pyLDAvis.gensim_models as gensimvis
pyLDAvis.enable_notebook()
C:\Anaconda3\lib\site-packages\sklearn\decomposition\onli
ne_lda.py:31: DeprecationWarning: `np.float` is a deprecat
ed alias for the builtin `float`. To silence this warning,
use `float` by itself. Doing this will not modify any behav
ior and is safe. If you specifically wanted the numpy scala
r type, use `np.float64` here.
Deprecated in NumPy 1.20; for more details and guidance: ht
tps://numpy.org/devdocs/release/1.20.0-notes.html#depreca
tions
  EPS = np.finfo(np.float).eps
```

In [80]:

```
import pyLDAvis.gensim_models
```

In [81]:

```
pyLDAvis.gensim_models.prepare(ldamodel, corpus_bow, dictionary)
C:\Anaconda3\lib\site-packages\pyLDAvis\_prepare.py:247:
FutureWarning: In a future version of pandas all arguments
of DataFrame.drop except for the argument 'labels' will be
keyword-only
  by='saliency', ascending=False).head(R).drop('saliency',
 1)
```

Out[81]:

In [82]:

```
ldamodel.show_topics()[0]
```

Out[82]:

```
(0,
 '0.041*"batch" + 0.038*"batches" + 0.031*"release" + 0.02
8*"DS" + 0.024*"lot" + 0.023*"data" + 0.023*"DP" + 0.020*"p
rocess" + 0.019*"CTM" + 0.018*"manufacturing"')
```

In [83]:

In [84]:

```
new_query = "Certificate of analysis from alternative supplier (Hongene Biotech Corporation) for GTP (3.2.S.2.3)"
```

In [85]:

```
from nltk.corpus import stopwords
```

In [86]:

```
new_query = re.sub("[()]","", new_query)
```

In [87]:

```
new_query
```

Out[86]:

```
'Certificate of analysis from alternative supplier Hongene
 Biotech Corporation for GTP 3.2.S.2.3'
```

In [88]:

```
tokenized_nq = new_query.split(' ')
```

In [89]:

```
tokenized_nq
```

Out[89]:

```
['Certificate',
 'of',
 'analysis',
 'from',
 'alternative',
 'supplier',
 'Hongene',
 'Biotech',
 'Corporation',
 'for',
 'GTP',
 '3.2.S.2.3']
```

In [90]:

```
tokenized_nq = [word for word in tokenized_nq if word not in stopwords.words('english')]
```

In [91]:

```
tokenized_nq
```

Out[91]:

```
['Certificate',
 'analysis',
 'alternative',
 'supplier',
 'Hongene',
 'Biotech',
 'Corporation',
 'GTP',
 '3.2.S.2.3']
```

```
query_vector = dictionary.doc2bow(tokenized_nq)
```

In [92]:

```
from gensim import similarities
```

In [93]:

```
feature_cnt = len(dictionary.token2id.keys())
```

In [94]:

```
index = similarities.SparseMatrixSimilarity(tfidf[corpus_bow], num_features=f
eature_cnt)
```

In [95]:

```
sim = index[tfidf[query_vector]]
```

In [96]:

```
result_list = []
```

In [97]:

```
print('\n''\n')
print('Searched query:', new_query)
print('\n')
for i in range(len(sim)):
    if sim[i] > 0.5:
        result_list.append(ner_corpus[i])
        print('similarity with query %d: %.2f' % (i + 1, sim[i]))
        print(ner_corpus[i])
        print('\n')
```

```
Searched query: Certificate of analysis from alternative s
upplier Hongene Biotech Corporation for GTP 3.2.S.2.3


similarity with query 2: 0.53
batch analysis commercial-scale batches alternative Nucleo
side Triphosphates GTP Hongene Biotech Corporation data 3.
2.S.4.4


similarity with query 4: 0.89
Certificate of analysis alternative supplier Hongene Biote
ch Corporation GTP


similarity with query 408: 0.71
CoA GTP supplier Hongene Biotech Corporation
```

similarity with query 1095: 0.54
details information material GTP alternative Hongene Biotech Corporation synthesis material materials reagents solvents
b. Exemplary batch certificates batch material Hongene Biotech Corporation